Homework 7: Planar bifurcations and periodic orbits

MATH 515 — Fall 2026

Due Friday, December 4, 2026

Problem 1: A planar saddle-node

Consider

$$\dot{x} = \mu - x^2, \qquad \dot{y} = x - 2y.$$

(a) Find all equilibria and classify their linearizations as μ varies. On the branch with $x > 0$, determine when the two eigenvalues coincide. At that parameter, give the dimension of the eigenspace and say whether the equilibrium is hyperbolic.

(b) At $\mu = 0$, determine whether the origin is Lyapunov stable. Use the equation for x to justify your answer.

Problem 2: Existence in a trapping annulus

Consider the system

$$\dot{x} = \frac{1}{4}(2 - x^2 - y^2)x - y, \qquad \dot{y} = x + \frac{1}{4}(1 - x^2 - y^2)y.$$

(a) Write the equations for $\rho = \sqrt{x^2 + y^2} > 0$ and θ in polar coordinates. Show that $\dot{\theta} > 0$ and that the annulus $9/10 \le \rho \le 3/2$ is forward invariant.

(b) Use the Poincaré–Bendixson theorem to show that this annulus contains a periodic orbit.

(c) Compute the divergence. A student uses its sign on the annulus to argue that the Bendixson criterion rules out the periodic orbit from part (b). Explain where that argument goes wrong.

Problem 3: Hopf bifurcation and the first-return multiplier

Consider the system

$$\dot{x} = \big(\mu - 2(x^2+y^2)\big)x - (2 + x^2 + y^2)y, \qquad \dot{y} = (2 + x^2 + y^2)x + \big(\mu - 2(x^2+y^2)\big)y.$$

(a) Find the eigenvalues at the origin. Classify the origin for $\mu < 0$ and $\mu > 0$. At $\mu = 0$, use the nonlinear equations to determine its stability.

(b) Find the nonconstant periodic orbits and classify their radial stability. Identify the bifurcation at $\mu = 0$.

(c) When a periodic orbit exists, find its period and the derivative of the first-return map at the orbit's intersection with the positive x-axis. Use the radius as the coordinate on that section. For the derivative, take θ as the independent variable and linearize the resulting radial equation about the cycle.

Problem 4: Measuring the return map [Challenge]

Consider the system

$$\dot{\rho} = 2 + \cos\theta - \rho, \qquad \dot{\theta} = 1, \qquad \rho > 0.$$

Pick any radius $\rho_0 > 0$ and start on the positive x-axis, so $\rho(0) = \rho_0$ and $\theta(0) = 0$. Follow the trajectory until it next crosses the positive x-axis. Let $P(\rho_0)$ denote its radius when it returns. Trying a different starting radius gives another value of the same map P.

(a) Solve for $\rho(t)$ and show that it stays positive. When does the trajectory first return to the positive x-axis? Use this time to find a formula for $P(\rho_0)$.

(b) Is there a starting radius for which the trajectory comes back to the same point? Find it, write down the corresponding periodic solution, and give its least period.

(c) Find the radius after n returns and determine its limit as $n \to \infty$. What happens between returns? Is the periodic orbit attracting?